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Daniel Precioso, PhD

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Data Scientist

7+ years delivering ML/DL systems across healthcare, energy, insurance, and maritime logistics. I combine technical rigor (PhD, 5 Q1 publications) with strong technical communication, presenting model outputs to executives, building production applications, and designing clear materials for clients and public audiences.

Technical Skills

Languages	:	Python (expert), R, MATLAB, SQL
ML/DL	:	scikit-learn, TensorFlow, PyTorch, Keras, JAX
GenAI & LLMs	:	OpenAI API, Claude, Gemini, Multi-agent systems
MLOps & DeVOps	:	Git (CI/CD, pre-commit hooks), Flask, Docker (basics)
Cloud & Data	:	AWS S3, IAM, Pandas, NumPy, Streamlit
Academic Tools	:	LaTeX, Overleaf, Blackboard, WordPress

Experience

Postdoctoral Researcher & Adjunct Professor — Sep 2023 – Present
IE University — *Madrid, Spain*

- Work on climate-mortality research for Vienna Insurance Group (Phase 1 team member, Phase 2 lead): present to the steering committee and board, deploy a Flask application (PythonAnywhere), and implement CI/CD workflows. Built climate-mortality models (DLNMs, Relative Risk curves).
- Implemented Kolmogorov-Arnold Networks (KANs) for weather forecasting, improving prediction accuracy.
- Built and maintained IE Research Datalab's website, increasing research visibility and collaboration opportunities.
- Delivered technical courses to 100+ students with consistently high satisfaction ratings.

COO — Apr 2023 – Present
Canonical Green — *Madrid, Spain*

- Built route optimization models for maritime clients; in pilot simulations with a tugboat operator, weather routing cut GHG emissions by about 10%.
- Managed large-scale **AWS S3** datasets, configured access permissions, and maintained cloud-to-on-prem data transfer workflows.
- Prepared and delivered technical presentations in pitches, conferences, and client meetings, translating routing algorithms into clear operational and business value.
- Built and maintained startup communication assets, including website updates, news content, and talk materials that increased visibility among potential clients and collaborators.

Data Scientist — Sep 2022 – Apr 2023
Komorebi AI — *Madrid, Spain*

- Cleaned and processed 500K+ data points from raw sensor data, enabling accurate fish population monitoring.
- Built gradient-boosted and CNN models to estimate tuna biomass under echosounder buoys.
- Built an interactive dashboard used by the operations team.

Predoctoral Research Staff Sep 2019 – Aug 2022
University of Cádiz *Cádiz, Spain*

- Built ML models for Industry 4.0 applications in energy monitoring and non-destructive testing, reducing operational costs for industrial partners.
- Partnered with Spain's Health Alerts Centre on COVID-19 ICU forecasting models.
- Published 5 peer-reviewed papers in Q1/Q2 journals, advancing the field of applied ML and DS.

Junior Data Scientist Jan 2019 – Jun 2019
Foqum *Madrid, Spain*

- Built neural network models using Keras and TensorFlow for non-intrusive load monitoring (NILM) in energy sector.

EDUCATION

PhD in Computer Engineering Sep 2019 – Jul 2023
University of Cádiz *Cádiz, Spain*

MSc in Statistical and Computational Information Processing Sep 2018 – Jul 2019
Universidad Politécnica de Madrid *Madrid, Spain*

Degree in Physics Sep 2014 – Jul 2018
Complutense University of Madrid *Madrid, Spain*

PROJECTS

KAN4Met *Machine Learning, Meteorology* IE University

- Implemented Kolmogorov-Arnold Networks for weather forecasting, advancing ML-based prediction accuracy.
- Contributed to research team that secured funding from Spanish Ministerio de Ciencia, Educación y Universidades.

Insurance and Tech *Data Science, Actuarial Science* IE University

- Built models of the impact of climate on mortality, for use in life-insurance analysis.
- Contributed to industry-academic partnership with Vienna Insurance Group (VIG), translating complex climate models into actionable business insights.

Weather Navigation *Mathematical Optimization, Maritime Transport* IE University

- Designed optimization algorithms for weather routing; in pilot simulations they cut GHG emissions by about 10%.
- Team member in projects funded by the BBVA Foundation and the Spanish Agencia Estatal de Investigación (grant TED2021-129455B-I00).
- Built and maintained the official project website https://weathernavigation.com/.
- Presented technology to the European Parliament.

Tun-AI *Deep Learning, Fishing Industry* Komorebi AI

- Partnered with Satlink on machine-learning models for tuna monitoring.
- Transformed 500K+ raw echosounder readings into clean, structured datasets that powered ML models.
- Trained gradient-boosted trees and CNNs for fish monitoring from echosounder data.

NeoCam *Computer Vision, Healthcare* University of Cádiz

- Built a contactless monitoring system using Luxonis OAK-D cameras that enabled health experts to track newborn vitals in NICUs.
- Won 2nd place globally at OpenCV AI Competition 2021 among 1,400+ teams.
- Co-inventor on official utility-model registration in 2023 (utility model, Spain).

COVID-19 *Time Series Forecasting, Healthcare* University of Cádiz

- Partnered with Spain's Coordination Centre for Health Alerts to build ICU occupancy forecasting models.

- Analyzed non-pharmaceutical interventions (NPIs) impact on COVID-19 spread, providing evidence on which measures reduced transmission.

UCAnFly *Computer Modeling, Astrophysics* University of Cádiz

- Built MATLAB simulations modeling Earth's electromagnetic field effects on gravitational test mass for space-based detectors like LISA.
- Contributed electromagnetic modeling expertise to multidisciplinary team selected by ESA's Fly Your Satellite! programme.

ATENEA *Computer Vision, Industrial Automation* University of Cádiz

- Built ML and NLP solutions for Airbus D&S that automated production processes, reducing manual inspection time.
- Contributed to industry partnership funded by CDTI Interconnecta program.

Courses and Certifications

- AWS, AWS Cloud Practitioner Essentials.
- France Université Numérique, Machine learning in Python with scikit-learn.
- UCM, XV Modeling Week at UCM (Coordinator).
- ESA, Fly your Satellite - 3 CDR Virtual Workshop.
- DeepLearning.AI (Coursera), TensorFlow in Practice Specialization.
- University of Michigan (Coursera), Introduction to Data Science in Python.
- University of Michigan (Coursera), Applied Text Mining in Python.
- University of Michigan (Coursera), Applied Social Network Analysis in Python.
- University of Michigan (Coursera), Applied Machine Learning in Python.

Publications

- New analysis in the preliminary design for LNG and LPG ships. Marine Structures, Elsevier, 2025 (Q1).
- HADAD: Hexagonal A-Star with Differential Algorithm Designed for weather routing. Ocean Engineering, 2024 (Q1)
- Hybrid Search method for Zermelo's navigation problem. Computational and Applied Mathematics, 2024 (Q2)
- Aggregation dynamics of tropical tunas around drifting floating objects based on large-scale echo-sounder data. Marine Ecology Progress Series, 2023 (Q1)
- Effectiveness of non-pharmaceutical interventions in nine fields of activity to decrease SARS-CoV-2 transmission (Spain, September 2020-May 2021). Frontiers in Public Health, 2023 (Q1)
- Thresholding Methods in Non-Intrusive Load Monitoring. The Journal of Supercomputing, 2023 (Q2)
- NeoCam: An edge-cloud platform for non-invasive real-time monitoring in neonatal intensive care units. IEEE Journal of Biomedical and Health Informatics, 2023 (Q1)
- TUN-AI: Tuna biomass estimation with Machine Learning models trained on oceanography and echosounder FAD data. Fisheries Research, 2022 (Q2)